

ADAL: AI-Driven Analysis for Law

Final Year Project 1 Presentation

Group Members:

- Husnain Ali Arshad
- Abdullah Azeem
- Hasan Ali

Supervisor: Mr. Muhammad Almas

Department: Computer Science

Slide 2 — Project Vision & Objectives

Bridge the gap in Pakistan's legal research by providing an AI-powered system that simplifies legal research, drafting, and validation.

Core Functions:

- Document ingestion
- Citation verification
- Case summarization
- Precedent retrieval (Implemented using NLP and Retrieval-Augmented Generation)

Slide 3 — Iteration Focus & Module Status

Overall Project Modules (5):

1. Document Ingestion
2. Citation Verification
3. AI Summarization & Precedent Retrieval
4. Legal Assistant
5. UI & Reporting

Modules Completed (FYP-1 Scope):

- Module 1 — Document Ingestion & Preprocessing (PDF/Text/Image upload, OCR & text extraction)

Current Iteration Focus:

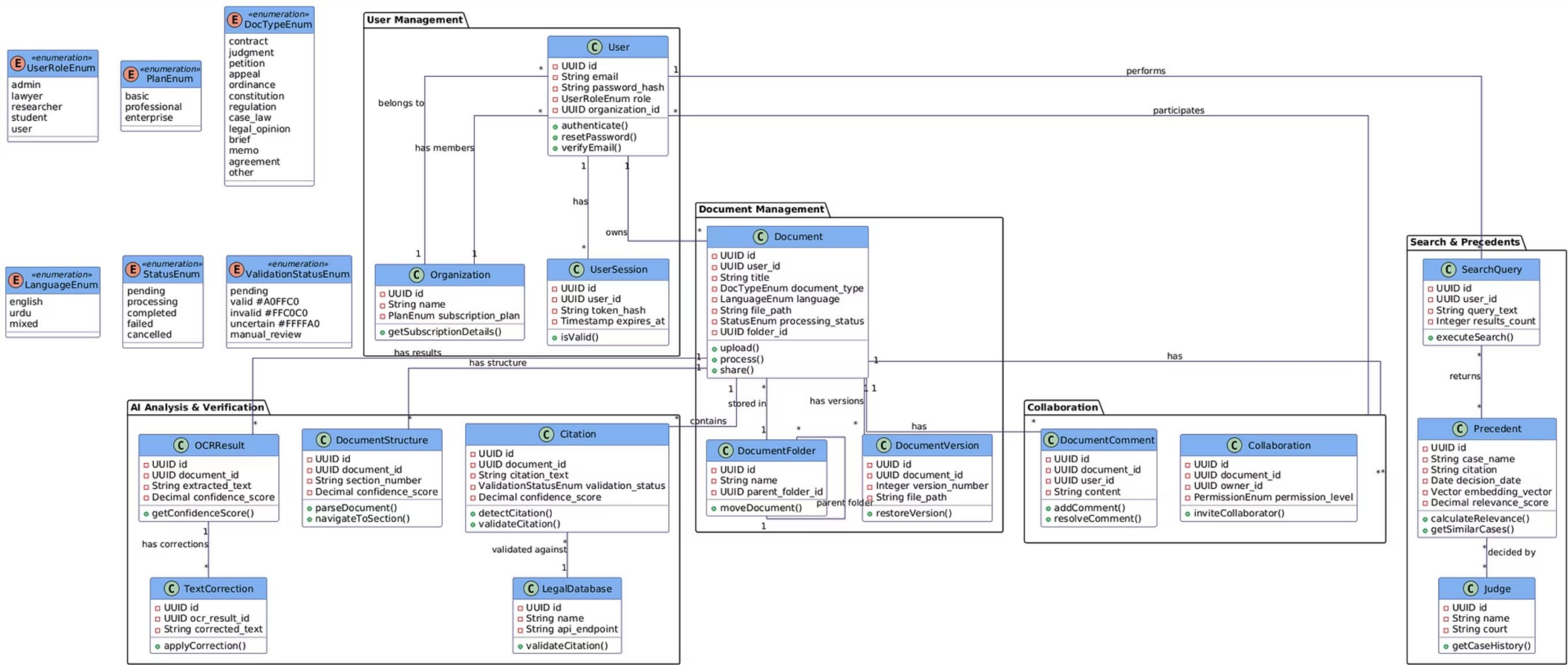
- Module 2 — Citation Verification
- Module 3 — AI Summarization & Precedent Retrieval

Slide 4 — Addressing Previous Feedback: Design Models

Improved clarity and level of detail in system design models.

Class Diagram:

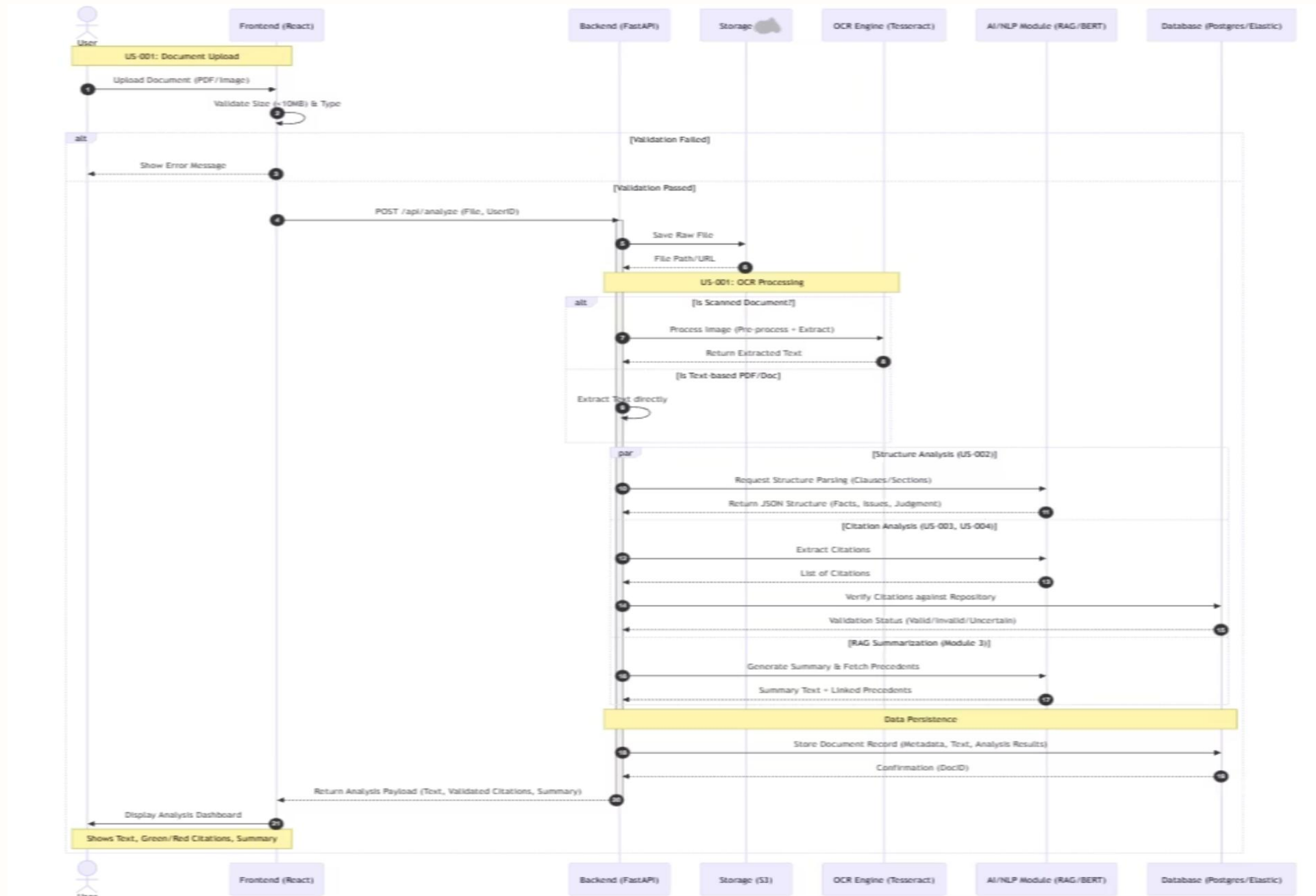
- Expanded backend entities and associations
- Shows class-level structure clearly



Slide 5

Sequence Diagram:

- Detailed interaction flow between: Frontend → FastAPI Backend → OCR Engine → AI/NLP Module



Slide 6 — Addressing Previous Feedback: Security

Security Focus: Login, authentication, and data protection

Functional Requirements:

- User registration and authentication
- Role-based access control

Non-Functional Security Measures:

- Access restricted to authenticated users only

Slide 7 — Module 2: Citation Verification

Automatically detect and validate legal citations against an internal judgments repository.

1

Key Algorithm:

- Citation Extraction & Validation Pipeline
- Rule-based + NLP pattern matching

2

Tools Used:

- spaCy / NLTK
- PostgreSQL / SQLite



Slide 8 — Module 3: AI Summarization & Precedent Retrieval

Use AI and RAG models to summarize legal cases.



1

Key Algorithm:

- Retrieval-Augmented Legal Draft Generation

2

Tools Used:

- Hugging Face Transformers (LLMs)
- FAISS / ChromaDB (Vector DB)
- Sentence Transformers (Embeddings)

Slide 9 — Conclusion & Next Steps

Successfully delivered core modules:

- Document Ingestion (M1)
- Citation Verification (M2)
- RAG-based AI Analysis (M3)

Next Steps (FYP-2):

- Integrate knowledge graph and semantic search
- Add caching and system optimizations
- Develop Legal Assistant and enhanced UI modules